

Badr Mouline

Montreal, QC | +1 (438) 270 3219 | badr.mouline@umontreal.ca | [LinkedIn](#) |

Professional profile

- **Computer Science Student | Software Engineering Aspirant**, with practical experience in **IT systems in government departments** (maintenance, networks, workstation upgrades, computerized management).
- Notable achievements: SpaceX predictive model (**IBM Data Science**), Cyclistes en R data analysis (Google Data Analytics), **OCR solution with Python**, Open CV, and Tesseract.
- Curious, rigorous, creative, and persistent, I enjoy working in teams in innovative environments.

Technical skills

- **Programming and Data Science:** Python, R, Java, SQL, HTML/CSS, C, JS,
- **Tools and software:** Office 365, VBA, Tableau, Power BI, Figma, Sage 100, Méga Compta, AWS, Git/GitHub, Maven, Junit, Figma, Tableau, Power BI.
- **Analysis and AI:** Machine learning, Deep learning, NLP, Neural networks, Data cleaning, Data management, Data collection, Data analysis.
- **Databases & CS concept:** SQL Server, Oracle, Access, Data Structures & Algorithms, Operations Research, Programming Paradigms.
- **Math:** Linear algebra, Integral calculus, Probability, Statistics.

Certifications

- | | |
|---|----------|
| • AWS Cloud Solutions Architect —Amazon Web Services | 24-Nov |
| • Google UX Design —Google for Pros | 24-Jun |
| • IBM Data Science —IBM | 24-Mar |
| • Google Data Analytics —Google for Pros | 24-Jan |
| • Python 3 Programming —University of Michigan | 24-April |
| • Introduction to Machine Learning —Duke University | 24-Oct |

Formation

- | | |
|--|----|
| • Bachelor's degree in computer science - Artificial Intelligence , University of Montreal, Since Fall, | 25 |
| • Professional Technician Diploma in Computerized Management , MIAGE Group, Rabat, Morocco, | 23 |
| • Professional Qualification Diploma in Data Entry , MIAGE Group, Rabat, Morocco | 21 |

Awards

- | | |
|--|---------|
| • Excellence Scholarship - University of Montreal (Talented Students). | 25-28 |
| • 2nd graduating class – Professional Data Entry Operator Diploma, MIAGE Group. | 23-Dec |
| • Valedictorian – Computer Management Technician Diploma, MIAGE Group. | 21-July |

Languages

French (fluent), **English** (intermediate), **Arabic** (fluent), **Japanese** (beginner)

Professional experience

IT Assistant (intern), Ministry of Economy and Finance, Rabat, Morocco,

23 Oct-Dec

- Maintenance and troubleshooting of IT equipment and networks:
 - Quickly diagnose and resolve network and hardware issues, reducing service interruptions by 30%.
 - Collaborated with the team to modernize workstations while improving staff productivity
- Data entry and administrative management of legal cases:
 - Entered criminal and legal files and verified information before

Accounting Clerk (intern), IT service company, Rabat Morocco,

23 July-Nov

- Software development with Sage 100 and Mega Compta:
 - Entered invoices, verified balances (debit and credit balances).
- Preparing pay slips and monthly balance sheets, ensuring tax compliance and accuracy of financial data:
 - Transferring net pay to employees, verifying all current and non-current company expenses for the creation of monthly balance sheets.

Projects

Full-Stack Software Engineering Project – MaVille (Smart City App):

- **Architected and developed** a Java-based system to coordinate roadworks between residents, city agents, and contractors using **UML modeling** and the **C4 Architecture model**.
- **Integrated Montreal Open Data API** to provide real-time roadwork tracking and automated notification systems for citizens.
- **Implemented a robust testing suite** using **JUnit** and **Mockito**, ensuring high code reliability and coverage measured **via JaCoCo**.
- **Managed full development lifecycle** using **Git/GitHub** and **Maven**, following Agile principles for iterative delivery and client feedback.

IBM Data Science – Capstone Project: Space Y vs SpaceX:

- **Designing a predictive model with ML** to estimate the cost and reuse of rockets.
- **Collecting and cleaning** complex datasets while ensuring the reliability of results.
- **Presenting findings in an interactive dashboard**, a model for the space industry.

Google Data Analytics – Cyclist Case Study:

- Completed a comprehensive project in R: data collection, processing, visualization, and interpretation.
- Identified trends in cyclist performance and safety.
- **Applied advanced statistical techniques** to validate results and ensure their robustness.

SantéHub – Interactive dashboard for tracking habits – University of Montreal:

- **Designed and developed** a responsive single-page health-habit tracking web app from a Figma mockup **using pure HTML5 / CSS3 / JavaScript / JQuery / DOM**.
- Built interactive dashboards with dynamic charts and filters and improved accessibility and cross-browser support to ensure an inclusive user experience.

Areas of interest

Machine Learning, News, Robotic, Basketball, Sciences